

Power BI Basics

Introduction to Power BI

Power BI is a Business Intelligence (BI) and Data Visualization tool developed by Microsoft. It enables users to connect to multiple data sources, transform raw data into meaningful information, and create interactive dashboards and reports. Power BI helps organizations analyze data efficiently, identify trends, and make informed business decisions.

Power BI is widely used in industries such as finance, healthcare, education, retail, manufacturing, and information technology for reporting and data analysis.

Features of Power BI

- Connects to multiple data sources such as Excel, CSV, SQL Server, Oracle, SharePoint, Azure, and cloud services.
- Cleans and transforms data using Power Query Editor.
- Creates interactive dashboards and reports.
- Supports drag-and-drop report development.
- Uses DAX (Data Analysis Expressions) for advanced calculations.
- Allows data modeling and relationship creation.
- Provides real-time dashboard updates.
- Enables report sharing through Power BI Service.
- Supports mobile applications for Android and iOS.
- Offers AI-powered insights and forecasting.

Components of Power BI

1. Power BI Desktop

Power BI Desktop is a free Windows application used for importing data, transforming data, building data models, creating visualizations, and designing reports.

Uses

- Data import
- Data cleaning
- Report creation
- Dashboard design
- Data modeling

2. Power BI Service

Power BI Service is the cloud-based version of Power BI. It is used for publishing reports, sharing dashboards, collaborating with team members, and scheduling data refresh.

Uses

- Publish reports
- Share dashboards
- Schedule refresh
- Collaboration
- Workspace management

3. Power BI Mobile

Power BI Mobile allows users to access dashboards and reports from smartphones and tablets anytime and anywhere.

4. Power BI Gateway

Power BI Gateway securely connects on-premises data sources with the Power BI cloud service for automatic data refresh.

Power BI Workflow

The general workflow of Power BI is:

1. Connect to data sources.
2. Load or transform data.
3. Build relationships between tables.
4. Create calculated columns and measures.
5. Design charts and visualizations.
6. Create reports.
7. Build dashboards.
8. Publish reports to Power BI Service.
9. Share reports with users.

Data Sources Supported by Power BI

Power BI supports numerous data sources, including:

- Microsoft Excel
- CSV Files
- Text Files
- SQL Server
- Oracle Database
- MySQL
- PostgreSQL
- Azure SQL Database

- SharePoint
- Google Analytics
- Salesforce
- Web APIs
- Microsoft Access

Power BI Interface

The Power BI Desktop interface mainly consists of:

Home Ribbon

Contains commands for importing data, transforming data, refreshing reports, and publishing reports.

Report View

Used to create charts, graphs, and interactive reports.

Data View

Displays imported data in tabular format.

Model View

Used to create relationships between multiple tables.

Fields Pane

Displays all tables, columns, and measures.

Visualizations Pane

Contains all available charts and formatting options.

Filters Pane

Used to filter report data at visual, page, or report level.

Types of Visualizations in Power BI

Power BI offers various visualization options, including:

- Bar Chart
- Column Chart
- Line Chart

- Area Chart
- Pie Chart
- Donut Chart
- Tree Map
- Scatter Plot
- Waterfall Chart
- Funnel Chart
- Gauge Chart
- Card
- Multi-row Card
- KPI
- Table
- Matrix
- Map
- Filled Map
- Ribbon Chart
- Decomposition Tree
- Key Influencers
- Slicer

Data Transformation Using Power Query

Power Query Editor is used to clean and prepare data before analysis.

Common transformations include:

- Remove duplicate values
- Replace missing values
- Split columns
- Merge columns
- Append tables
- Merge queries
- Change data types
- Filter rows
- Sort data
- Group data
- Pivot and Unpivot columns

Data Modeling

Data Modeling organizes multiple tables into a structured model.

Important concepts include:

- Primary Key
- Foreign Key
- One-to-One Relationship
- One-to-Many Relationship
- Many-to-Many Relationship

- Active Relationship
- Inactive Relationship
- Star Schema
- Snowflake Schema

DAX (Data Analysis Expressions)

DAX is the formula language used in Power BI to perform calculations and create measures.

Common DAX Functions:

Aggregate Functions

- SUM()
- AVERAGE()
- MIN()
- MAX()
- COUNT()

Logical Functions

- IF()
- AND()
- OR()

Date Functions

- TODAY()
- YEAR()
- MONTH()
- DAY()

Text Functions

- LEFT()
- RIGHT()
- CONCATENATE()

Measures vs Calculated Columns

Calculated Column

- Calculated for every row.
- Stored in memory.
- Increases model size.
- Used for row-level calculations.

Example:

Total Marks = Maths + Science + English

Measure

- Calculated only when required.
- Not stored in memory.
- Faster and more efficient.
- Used for aggregations.

Example:

Average Marks = AVERAGE(Student[Marks])

Filters in Power BI

Power BI provides three types of filters:

Visual-Level Filter

Applies only to one visualization.

Page-Level Filter

Applies to all visuals on a single report page.

Report-Level Filter

Applies to every page in the report.

Slicers

A Slicer is an interactive filter that allows users to dynamically filter report data.

Examples:

- Year
- Department
- Gender
- Country
- Product Category

Dashboard vs Report

Dashboard

- Single page
- Combines multiple visuals
- Interactive
- Created in Power BI Service

Report

- Multiple pages
- Detailed analysis
- Created in Power BI Desktop
- Contains multiple visualizations

Applications of Power BI

Power BI is used in various domains:

- Business Intelligence
- Sales Analysis
- Financial Reporting
- Marketing Analytics
- Human Resource Analytics
- Inventory Management
- Healthcare Reporting
- Education Analytics
- Manufacturing Analysis
- Customer Behavior Analysis

Conclusion

Power BI is one of the most popular business intelligence and data visualization tools available today. It enables users to transform raw data into meaningful insights through interactive dashboards, reports, and visual analytics. With features such as data connectivity, Power Query, DAX, data modeling, and cloud-based collaboration, Power BI has become an essential tool for organizations seeking data-driven decision-making. Learning the fundamentals of Power BI provides a strong foundation for developing advanced analytics and reporting solutions.

You can also expand this into a **25–30 page professional document** by adding screenshots of the Power BI interface, examples of each visualization, Power Query demonstrations, DAX examples, and dashboard design best practices.

